

Class Test: Mathematics for Computing

Answer all the questions

Full Marks: 30

Time: 60 minutes

1. ~~a.~~ Let $W = \{(x, y, z) \in \mathbb{R}^3 : x - 4y + 3z = 0\}$. Show that W is a subspace of the real vector space $\mathbb{R}^3$.
Find the dimension of W .
 - b. Show that the three vectors $v_1 = (0, 3, 1, -1)$, $v_2 = (6, 0, 5, 1)$, and $v_3 = (4, -7, 1, 3)$ form a linearly dependent set in $\mathbb{R}^4$.
 - c. Define Spanning Set and Basis of a vector space. Is there any relation between spanning set and basis of a vector space? What do you mean by Orthonormal basis?
 - d. Use row reduction to solve the linear system.

$$\begin{aligned} 2x_1 + 4x_2 + 6x_3 &= 8 \\ x_1 + 2x_2 + 4x_3 &= 8 \\ 3x_1 + 6x_2 + 9x_3 &= 12 \end{aligned}$$
- (3+1)+3+(2+1+1)+4=15
2. a. Prove that if $\{a_n^{(p)}\}$ is a particular solution of the nonhomogeneous linear recurrence relation with constant coefficients $a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k} + F(n)$, then every solution is of the form $\{a_n^{(p)} + a_n^{(h)}\}$, where $\{a_n^{(h)}\}$ is a solution of the associated homogeneous recurrence relation $a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k}$.
 - b. Find the generating function for the solutions to $h_n = h_{n-1} + h_{n-2}$, $h_0 = 1$, $h_1 = 3$, and use it to find a formula for h_n .
 - c. Find the solution of the recurrence relation $a_n = 4a_{n-1} - 3a_{n-2} + 2^n + n + 3$ with $a_0 = 1$ and $a_1 = 4$.
- 6+4+5=15